

HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

Final Test

- Time: 60 minutes Date: _____
- Closed book examination: NO Laptop, NO phone

Full name: _____ Student's number: _____

ANSWER SHEET

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

(Record your answers on the answer sheet)

1. Computer science is best described as
 - A. Writing computer programs only
 - B. Using computers for office work
 - C. The study of computation, algorithms, hardware, and software
 - D. Manufacturing electronic devices

2. A CPU has a 36-bit address bus. If each memory location stores 1 byte, what is the maximum addressable memory?
 - A. 32 GB
 - B. 64 GB
 - C. 128 GB
 - D. 256 GB

3. Software that controls hardware and provides a platform for applications is called
 - A. Application software
 - B. Utility software
 - C. System software
 - D. Firmware

4. What are the values of A and B after executing the following pseudocode?
 $A \leftarrow 5$, $B \leftarrow 3$
If ($A > B$) then
 $A \leftarrow A + B$
 $B \leftarrow A - B$
End if
 - A. A=5, B=3
 - B. A=8, B=5
 - C. A=8, B=3
 - D. A=3, B=5

5. Which component temporarily holds instructions and data currently being processed by the CPU?
 - A. Hard disk
 - B. Main memory (RAM)
 - C. Cache memory
 - D. Optical disk

6. A system of interconnected computers that share resources is called
 - A. Internet
 - B. Network
 - C. Server
 - D. Database

7. The main communication pathway that transfers data between components in a computer is the
 - A. Register
 - B. Bus
 - C. Port
 - D. Socket

8. Transaction Processing Systems (TPS) mainly
 - A. Support long-term strategic decisions
 - B. Process routine daily business transactions
 - C. Replace management decision-making
 - D. Store multimedia data

9. Which memory is non-volatile?
 - A. Cache
 - B. RAM
 - C. ROM
 - D. Register

10. Which statement about operating systems is incorrect?

- A. OS manages CPU, memory, and I/O devices
- B. OS provides an interface between user and hardware
- C. Open-source OS cannot be modified
- D. OS controls program execution

11. Given two 8-bit signed integers: $A = +60$ and $B = -20$. What is the result of $A + B$?

- A. +40
- B. +80
- C. -40
- D. -80

12. The lowest-level programming language is

- A. High-level language
- B. Assembly language
- C. Machine language
- D. Procedural language

13. Given an 8-bit unsigned integer $A = 11001010_2$, what is $\text{NOT}(A)$?

- A. 00110101
- B. 00110110
- C. 11001010
- D. 10110011

14. Convert the binary number 11010111.1010 to hexadecimal

- A. D7.A
- B. C7.B
- C. D6.A
- D. E7.A

15. If $(7F)_{16} = (X)_{10}$, the value of X is

- A. 112
- B. 127
- C. 135
- D. 255

16. The binary equivalent of hexadecimal number A3 is

- A. 10100011
- B. 10010011
- C. 10110001
- D. 11000011

17. The decimal equivalent of binary number 11101001 is

- A. 225
- B. 233
- C. 235
- D. 241

18. Software that allows users to create tables with rows and columns and perform calculations is

- A. Word processor
- B. Presentation software
- C. Spreadsheet
- D. Database

19. In a relational database, each row in a table represents a

- A. Field
- B. Attribute
- C. Record
- D. Key

20. Which of the following is not application software?

- A. Microsoft Word

- B. Google Chrome
- C. Linux Kernel
- D. Microsoft Excel

21. Which statement about I/O devices is correct?

- A. All I/O devices connect directly to main memory
- B. I/O devices communicate with CPU through I/O interfaces
- C. I/O devices do not need drivers
- D. I/O devices are part of the CPU

22. Software distributed with source code available for modification is

- A. Proprietary software
- B. Commercial software
- C. Open-source software
- D. Trial software

23. An OS that allows multiple programs to reside in memory and execute concurrently is

- A. Batch OS
- B. Single-user OS
- C. Multitasking OS
- D. Embedded OS

24. Given $X = 1010_2$, $Y = 12_8$, $Z = 9_{16}$. Arrange them in ascending order

- A. X, Z, Y
- B. Z, X, Y
- C. Y, X, Z
- D. Z, Y, X

25. A program that translates source code line by line during execution is called a/an

- A. Compiler
- B. Interpreter
- C. Assembler
- D. Emulator